

Global Mortality Patterns and Country Clustering: A Data-Driven Analysis of Causes of Death (2010–2019)

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Abstract Understanding what people die from, how mortality changes over time, and how mortality differs across countries is important for evidence-based public health planning. This study presents a data-driven analysis of global causes of death across countries from 2010 to 2019 using a country-year mortality dataset. The analysis examined overall mortality trends, leading causes of death, rate-based differences across income groups, latent mortality structures, clustering of countries with similar mortality profiles, and supervised model performance.

The analysis followed a structured workflow. The dataset was inspected and cleaned by checking for missing values, duplicate rows, duplicate country-year combinations, and invalid population values. Cause-specific death rates per 100,000 population were computed to standardize mortality across countries with different population sizes. Because the rate variables were strongly right-skewed, log transformation was applied. Exploratory data analysis was used to investigate global total deaths, average death rates, population trends, leading causes of death, and burden differences across income groups. Correlation analysis was used to examine relationships between demographic variables and major mortality causes. Principal Component Analysis (PCA) was then performed on standardized log-transformed mortality rates, followed by K-means clustering in PCA space. Finally, Linear Regression, Random Forest Regression, and Random Forest Classification were evaluated.

The results show that global total deaths increased over the study period, while the average death rate per 100,000 declined, indicating that mortality burden rose in absolute terms mainly because of population growth rather than worsening mortality conditions. Cardiovascular diseases and neoplasms were the dominant global causes of death. High-income countries showed a stronger chronic disease burden, whereas low-income countries showed higher infectious and maternal-neonatal burdens. PCA revealed two dominant mortality dimensions: an infectious/vulnerability burden and a chronic/lifestyle burden. K-means clustering identified three interpretable country profiles: high vulnerability burden, high chronic burden, and transition or mixed burden. Among the predictive models, Random Forest Classification performed best, achieving an accuracy of 98.4%, confirming that mortality cluster patterns were highly structured and predictable.

Index Terms global mortality, causes of death, epidemiological transition, PCA, clustering, random forest, classification, data mining

1. Introduction

Mortality is one of the most fundamental indicators of population health. It reflects not only biological disease burden but also the broader demographic, economic, and institutional conditions within which populations live. Patterns of death are shaped by age structure, healthcare access, sanitation, education, nutrition, disease exposure, economic development, and public policy. For this reason, mortality analysis provides a useful summary of both health status and social inequality [5, 6, 20].

A major idea in population health research is the epidemiological transition. In broad terms, this refers to the long-term shift from mortality dominated by infectious, maternal, neonatal, and nutritional conditions toward mortality dominated by chronic and noncommunicable diseases. However, this transition is uneven across countries. High-income countries typically show stronger burdens of cardiovascular disease, cancer, diabetes, and other chronic conditions. Many low-income countries continue to face high burdens from infectious diseases, diarrheal diseases, neonatal disorders, and other vulnerability-related conditions. Middle-income countries often display a mixed pattern in which both types of burden coexist [7–11].

This project was motivated by the need to examine these

differences using a structured data mining workflow rather than relying only on descriptive summary tables. A data mining approach allows the project to move from description to pattern discovery, variable reduction, cluster identification, and model-based validation. In this context, mortality is treated not simply as a collection of separate disease counts, but as a structured multivariate system that can be summarized, compared, and modeled [12, 14, 16].

The dataset used in this study contains country-year mortality information from 2010 to 2019. Since countries vary greatly in population size, raw death counts alone are not sufficient for fair comparison. Therefore, one of the most important preprocessing steps in this project is the conversion of cause-specific deaths into standardized rates per 100,000 population. This allows mortality to be studied as a comparable burden rather than merely a reflection of country size [1, 5, 11].

The study addresses the following questions:

- How did global mortality change from 2010 to 2019?
- Which causes contributed most to global deaths?
- How do mortality patterns differ across income groups?
- Can mortality variables be reduced into a smaller set of meaningful dimensions?

- Can countries be grouped into distinct mortality profiles?
- Which predictive model best captures the observed mortality structure?

By addressing these questions, the study provides both substantive public-health insight and a demonstration of how data mining methods can be applied to global mortality data.

2. Objective

The main objective of this study is to analyze global patterns in causes of death across countries from 2010 to 2019, examining temporal trends and differences across income groups.

The specific objectives are:

- To analyze global mortality trends from 2010 to 2019 using total deaths, death rates, and population trends.
- To identify the leading causes of death globally and examine how their contributions change over time.
- To compare mortality patterns across income groups using standardized cause-specific death rates and grouped burden categories.
- To investigate relationships between demographic variables and mortality outcomes using correlation analysis.
- To reduce the dimensionality of mortality variables using PCA and interpret the main mortality structures.
- To group countries into similar mortality profiles using K-means clustering.
- To evaluate predictive models and determine which model best represents the project objectives.

3. Related Literature

Previous public health research has consistently shown that mortality patterns vary with development level, demographic structure, and disease environment [5–8]. Studies on epidemiological transition have documented the shift from communicable to noncommunicable disease burden as countries industrialize and populations age. At the same time, work on global burden of disease has shown that this transition is incomplete and uneven, especially in countries that continue to face both chronic and infectious disease burdens [9–11, 21, 22].

Data-driven analyses of mortality increasingly use multivariate methods because disease variables are often correlated. Correlation analysis helps identify redundancy among disease indicators, while Principal Component Analysis provides a way to summarize large sets of mortality variables into a smaller number of interpretable dimensions. In applied health analytics, PCA is useful when several disease burdens move together and the analyst wants to identify the dominant underlying structure in the data [12, 13].

Clustering methods are also widely used in global health and development studies. Rather than focusing only on individual diseases, clustering can identify broad country

profiles that combine multiple mortality burdens. This is especially useful when countries do not fit neatly into a simple low-income versus high-income distinction. K-means clustering, when combined with PCA, provides a practical framework for grouping countries based on similar mortality signatures [14, 15].

Machine learning models have also been used to evaluate how predictable health structures are from multivariate patterns. In this project, supervised learning is not used to replace unsupervised analysis, but rather to validate the structure identified through PCA and clustering. This gives the study both an exploratory and a predictive dimension [16–19].

4. Dataset Description

The dataset consists of 1,920 observations and 40 variables. Each observation represents a country-year combination. The study period spans 2010 to 2019. The key variables include country name and code, income group, population, year, age 65+ percentage, dependency ratio, total deaths, overall death rate per 100,000, and cause-specific death counts.

The cause-of-death variables include, among others, cardiovascular diseases, neoplasms, chronic respiratory diseases, lower respiratory infections, diarrheal diseases, HIV/AIDS, tuberculosis, malaria, maternal disorders, neonatal disorders, diabetes mellitus, digestive diseases, chronic kidney disease, and cirrhosis.

Because the data are recorded at country-year level, the structure supports both temporal analysis and cross-country comparison. This makes the dataset suitable for trend analysis, income-group comparisons, multivariate dimension reduction, clustering, and supervised learning. It also allows the same country to be observed repeatedly over time, making it possible to examine both stability and change in mortality burden.

5. Methodology

5.1. Data Inspection

The dataset was loaded and its structure was reviewed, including row count, column count, variable names, and data types. This initial inspection was necessary to verify that the dataset contained the expected time coverage, population fields, and cause-of-death variables needed for rate computation and later analysis.

5.2. Data Quality Checks

The data was checked for missing values, full duplicates, duplicate country-year combinations, and invalid population values. These checks were essential because missing or duplicated observations would distort both descriptive trends and later multivariate analysis. Invalid population values would also make the standardized rate calculations unreliable. No missing values, duplicates, or invalid population entries were found.

5.3. Cause-Specific Rate Standardization

Cause-specific death rates per 100,000 population were computed as:

$$\text{Cause Rate per 100,000} = \left(\frac{\text{Cause Death Count}}{\text{Population}} \right) \times 100000 \quad (1)$$

This standardization step was critical because raw counts are strongly affected by country size. A country with a very large population may report many deaths even if its relative risk is low, while a smaller country may have fewer total deaths but a much higher mortality burden per capita. Standardized rates therefore provide a more appropriate basis for comparison across countries [1, 5, 11].

5.4. Distribution Checking and Log Transformation

The rate variables were strongly right-skewed, so log transformation was applied:

$$\log(1 + x) \quad (2)$$

This transformation was used because PCA and clustering are sensitive to highly skewed variables and outliers. By reducing skewness, the transformed variables become more suitable for multivariate analysis. The use of $\log(1 + x)$ also ensures that zero values can be handled safely [12, 13].

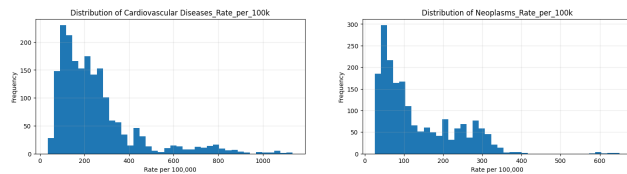


Figure 1: Selected mortality rate distributions before log transformation.

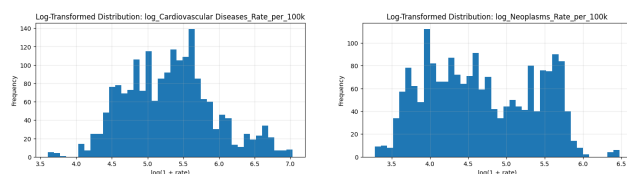


Figure 2: Selected mortality rate distributions after log transformation.

The histograms show that the original distributions are heavily right-skewed, with long upper tails caused by relatively few country-year observations with very high mortality rates. After transformation, the distributions become more compact and symmetric. This improves the stability of

downstream methods and reduces the dominance of extreme observations in PCA and clustering.

5.5. Global Trend Analysis

Global total deaths, average death rate per 100,000, and global population were plotted over time.

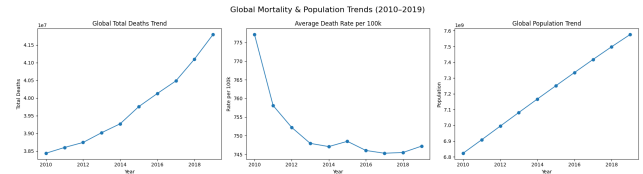


Figure 3: Global mortality, average death rate, and population trends from 2010 to 2019.

This step was included because total deaths and death rates answer different questions. Total deaths describe the absolute burden, while death rates describe relative burden after adjusting for population size. Including population trends makes it possible to interpret whether rising deaths are primarily due to worsening health outcomes or expanding population [1, 5, 10, 11].

5.6. Leading Cause Analysis

The top causes of death were aggregated and visualized using ranking and trend plots.

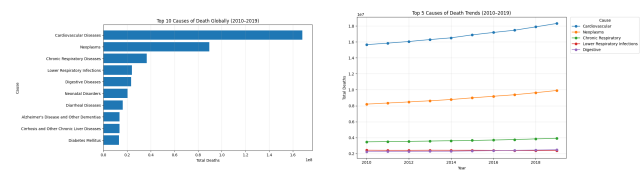


Figure 4: Top global causes of death and temporal trends of the top five causes.

This analysis identifies which causes dominate global mortality and whether the leading causes are stable over time. It also provides a descriptive bridge between the global totals and the more detailed multivariate analysis that follows [1, 5, 10, 11, 21].

5.7. Income Group Comparison

Average burden categories were compared across income groups.

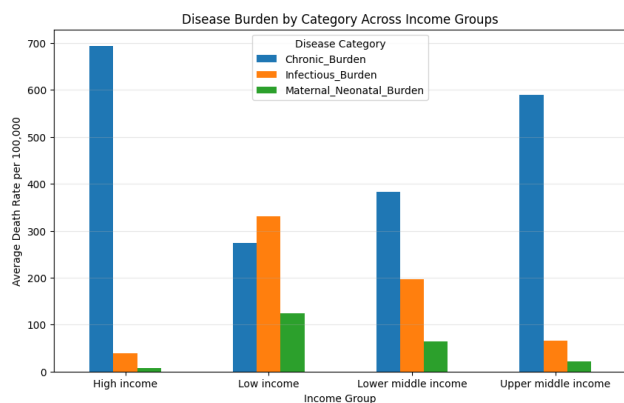


Figure 5: Average mortality burden categories across income groups.

To simplify interpretation, disease causes were grouped into broad categories such as chronic burden, infectious burden, and maternal-neonatal burden. This allows differences across income groups to be interpreted at a higher level than individual diseases alone [7–11].

5.8. Correlation Analysis

A heatmap was used to examine relationships among selected demographic and mortality variables.

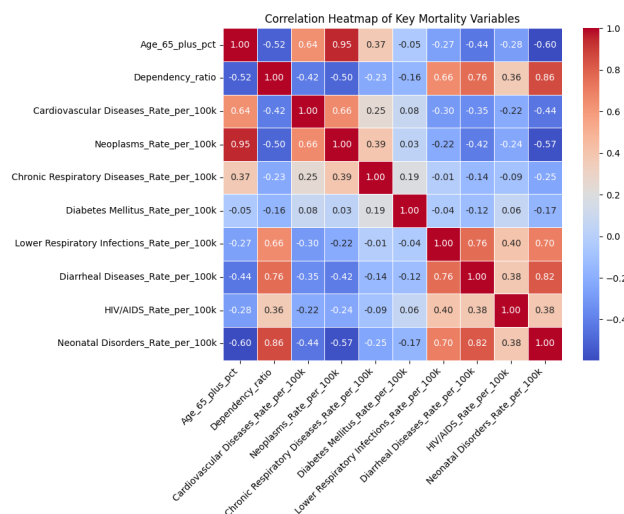


Figure 6: Correlation heatmap of key demographic and mortality variables.

Correlation analysis serves two purposes. First, it identifies substantive relationships among age structure, dependency ratio, and major disease categories. Second, it helps determine whether there is sufficient redundancy in the mortality variables to justify dimension reduction through PCA [12, 13].

5.9. Principal Component Analysis

PCA was applied to standardized log-transformed mortality rates.

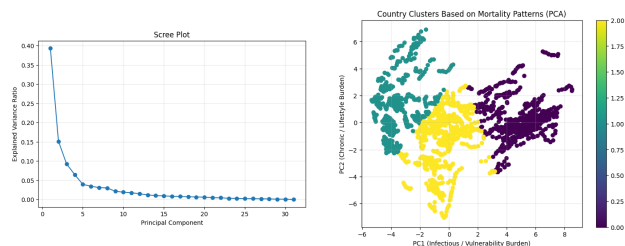


Figure 7: Scree plot and PCA projection of mortality patterns.

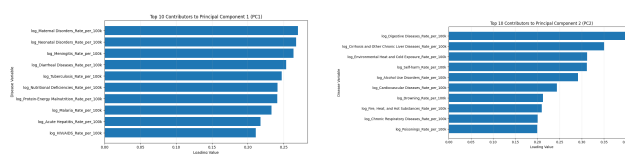


Figure 8: Top variable contributions to the first two principal components.

PCA reduces many correlated mortality variables into a smaller number of components that summarize the main structure of the data. The scree plot was used to assess how much variation is explained by the components, while the loading plots were used to interpret which disease variables contribute most strongly to each component [12, 13].

5.10. K-Means Clustering

The elbow method was used to select the number of clusters, and cluster burden profiles were then examined.

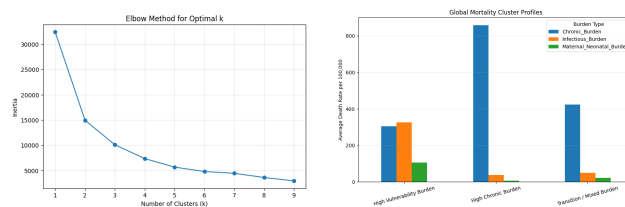


Figure 9: Elbow method for cluster selection and cluster burden profiles.

K-means clustering was applied in PCA space because the PCA coordinates provide a reduced and less redundant representation of mortality structure. The elbow method was used to guide the choice of the number of clusters, and

the resulting clusters were interpreted using their average chronic, infectious, and maternal-neonatal burdens [14, 15].

5.11. Geographic Mapping of Clusters

The dominant cluster membership was mapped by country.

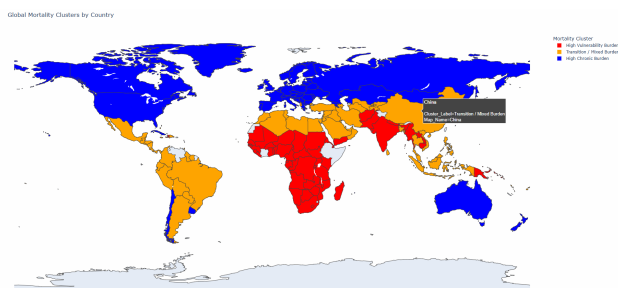


Figure 10: Geographic distribution of mortality clusters by country.

Geographic mapping makes the clustering results more interpretable by showing whether the mortality profiles align with recognizable regional or development-related patterns.

5.12. Representative Countries by Cluster

Countries with the highest mortality burden in each cluster were identified for interpretation.

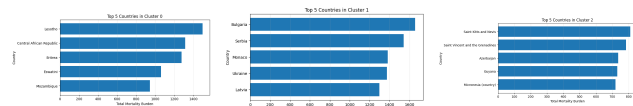


Figure 11: Top five countries by total mortality burden within each cluster.

This step was included to move from abstract cluster labels to concrete country examples. It helps confirm whether the clusters correspond to recognizable real-world mortality profiles.

5.13. Supervised Modeling

Three supervised models were tested:

- Linear Regression
- Random Forest Regression
- Random Forest Classification

Linear Regression and Random Forest Regression were used to predict cardiovascular disease mortality rates, while Random Forest Classification was used to predict mortality cluster membership. This part of the workflow extends the project beyond unsupervised learning and provides a direct comparison between simpler and more flexible predictive models [16–19].

6. Results

6.1. Global Mortality Trends

The global total deaths trend showed a steady increase from 2010 to 2019, while the average global death rate per 100,000 declined. Population increased steadily over the same period. Together, these patterns indicate that the increase in deaths was driven mainly by population growth rather than worsening mortality conditions. This distinction is important because rising total deaths alone could otherwise be misinterpreted as deteriorating health conditions [1, 5, 10, 11, 21].

6.2. Leading Causes of Death

Cardiovascular diseases were the leading cause of death globally, followed by neoplasms. Chronic respiratory diseases also ranked highly, while lower respiratory infections and diarrheal diseases remained important but less dominant than the major chronic conditions. This pattern indicates that chronic diseases account for a large share of overall global mortality burden [1, 5, 10, 11, 21].

The temporal trend plot shows that the top chronic causes remained consistently important across the study period. This suggests that the global mortality profile is not being driven by short-lived fluctuations but by persistent disease structures.

6.3. Income Group Differences

High-income countries showed the highest chronic burden, while low-income countries showed higher infectious and maternal-neonatal burdens. Lower-middle-income and upper-middle-income groups fell between these extremes, showing transition patterns rather than a complete dominance of one type of burden.

These results align with the idea of epidemiological transition. High-income settings are dominated by aging-related and lifestyle-related chronic diseases, while lower-income settings continue to face stronger communicable and vulnerability-related mortality. The middle-income groups reflect the coexistence of both types of burden [7–11].

6.4. Correlation Structure

The heatmap showed positive relationships between age 65+ percentage and chronic mortality variables such as neoplasms and cardiovascular disease. It also showed associations between dependency ratio and neonatal disorders, lower respiratory infections, and diarrheal diseases.

These patterns suggest that demographic structure is closely linked to mortality composition. Aging populations are associated with a higher chronic disease burden, while more vulnerable population structures are associated with infectious and maternal-neonatal burdens. The correlation patterns also confirm that some disease variables move together strongly enough to justify PCA [12, 13].

6.5. PCA Interpretation

The first principal component explained 39.3% of the variance and the second explained 15.1%, for a com-

Table 1: Summary of supervised model results

Model	Prediction Task	RMSE	R^2	Accuracy	Macro F1
Linear Regression	Cardiovascular disease rate prediction	133.70	0.44	–	–
Random Forest Regression	Cardiovascular disease rate prediction	47.23	0.93	–	–
Random Forest Classification	Mortality cluster classification	–	–	98.4%	0.98

bined 54.4%. The first component reflected infectious and vulnerability-related burden, while the second reflected chronic and lifestyle-related burden.

This means that much of the complexity in the original mortality variables can be summarized by two broad dimensions. Rather than interpreting dozens of disease variables separately, the analysis can describe countries in terms of where they lie on these dominant axes [12, 13].

6.6. Clustering Results

The elbow plot supported three clusters:

- Cluster 0: High vulnerability burden
- Cluster 1: High chronic burden
- Cluster 2: Transition or mixed burden

The cluster profiles confirmed that Cluster 0 had stronger infectious and maternal-neonatal burdens, Cluster 1 had the strongest chronic burden, and Cluster 2 represented a mixed or transitional profile. The PCA scatter plot showed visible separation among these groups, indicating that the cluster structure was not arbitrary [14, 15].

The mapped cluster distribution added geographic meaning to these patterns. Countries in Sub-Saharan Africa and some parts of South Asia were more concentrated in the high-vulnerability cluster, while Europe, North America, and other more developed settings were more concentrated in the high-chronic cluster.

6.7. Model Results

The performance of the three supervised models is summarized in Table 1.

Linear Regression provided only moderate explanatory power, indicating that a simple linear structure did not fully capture the relationships in the mortality data. Random Forest Regression performed much better, suggesting that nonlinear interactions and more flexible decision rules were important in predicting cardiovascular mortality rates. Random Forest Classification achieved the strongest overall performance, with an accuracy of 98.4% and a macro-average F1-score of 0.98, indicating that the cluster structure was highly separable and predictable [16–19].

7. Discussion

The study shows that global mortality patterns are shaped by a combination of demographic structure, development level, and disease transition. The coexistence of rising total deaths and declining death rates demonstrates why raw totals alone can be misleading. When population growth is taken into

account, the relative burden has declined even though the absolute number of deaths has increased [1, 5, 10, 11, 21].

The dominance of cardiovascular diseases and neoplasms confirms the central role of chronic disease in global mortality. However, the income-group analysis makes it equally clear that infectious and maternal-child burdens remain important in lower-income settings. The global picture is therefore not one of complete replacement, but one of uneven transition [7–11].

The PCA results show that mortality data contain strong latent structure. The first two components were able to summarize more than half of total variance, and their interpretation was substantively meaningful. This is important because it demonstrates that the complex mortality landscape can be reduced into a smaller set of interpretable dimensions without losing the main structure [12, 13].

The clustering results reinforce this conclusion. Countries do not appear randomly distributed in the mortality space; instead, they form interpretable groups that correspond broadly to vulnerability burden, chronic burden, and transitional burden. The world map and representative-country plots further support the practical meaning of these clusters [14, 15].

The supervised learning results add a validation layer to the project. While Random Forest Regression showed that continuous mortality outcomes can be predicted with high accuracy, Random Forest Classification was more directly aligned with the project’s core objective of validating mortality profiles. Its strong performance indicates that the clusters identified through unsupervised analysis are not weak or unstable patterns [16–19].

8. Practical Implications

The results have several practical implications. First, they show that global mortality patterns should not be interpreted using raw totals alone. Population-standardized rates provide a more meaningful basis for comparison across countries [1, 5, 11].

Second, the income-group findings suggest that health priorities should differ across development contexts. High-income countries may need stronger emphasis on chronic disease prevention, aging populations, and long-term management, while low-income countries may still require substantial investment in infectious disease control, maternal care, and neonatal health [5, 10, 11, 14].

Third, the clustering results show that countries can be grouped into broader mortality profiles. This may be useful for comparative policy analysis, regional benchmarking, or targeting interventions toward countries with similar burden structures rather than relying on income labels alone [14, 15].

Finally, the strong model performance suggests that mortality structures are sufficiently patterned to support predictive analytics. This does not replace public health judgment, but it does indicate that data mining methods can support structured comparison and decision-making [16, 18, 19].

9. Limitations

This study has several limitations. First, the analysis is based on country-year aggregated data rather than individual-level

observations. As a result, the findings describe population-level patterns and should not be interpreted as individual-level causal relationships.

Second, the study period is limited to 2010–2019. This provides a meaningful decade-long window, but it does not capture mortality changes after 2019, including the effects of the COVID-19 pandemic. Therefore, the findings should be interpreted as pre-pandemic mortality patterns.

Third, while the standardized rates improve comparability across countries, they do not eliminate all data-quality differences that may exist across international sources. Differences in reporting, diagnostic practice, and data completeness may still influence the results [5, 10, 11].

Fourth, the supervised models were used here primarily for pattern validation rather than causal explanation. Strong predictive performance should therefore not be interpreted as proof of causal relationships among mortality variables [16, 18, 19].

10. Conclusion

This study analyzed global mortality patterns across countries from 2010 to 2019 using a complete data mining workflow. The results showed that total deaths increased globally, death rates declined, chronic diseases dominated overall mortality, and countries could be grouped into three meaningful mortality profiles.

The analysis also showed that mortality patterns differ clearly across income groups, that a large share of variation can be summarized by two principal components, and that the resulting mortality profiles can be predicted with very high accuracy. These results suggest that global mortality patterns are structured, development-linked, and suitable for analysis using data mining methods [7, 10, 12, 14, 16].

Overall, the project demonstrates that combining preprocessing, standardization, exploratory analysis, PCA, clustering, and supervised learning provides a coherent framework for understanding global mortality differences.

Future Scope

Future work can extend this study by:

- incorporating socioeconomic predictors such as GDP, education, healthcare spending, and environmental factors
- using population-weighted mortality summaries
- exploring more advanced machine learning models
- extending the analysis into forecasting
- validating the findings using external datasets
- examining regional sub-patterns within continents and income groups

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